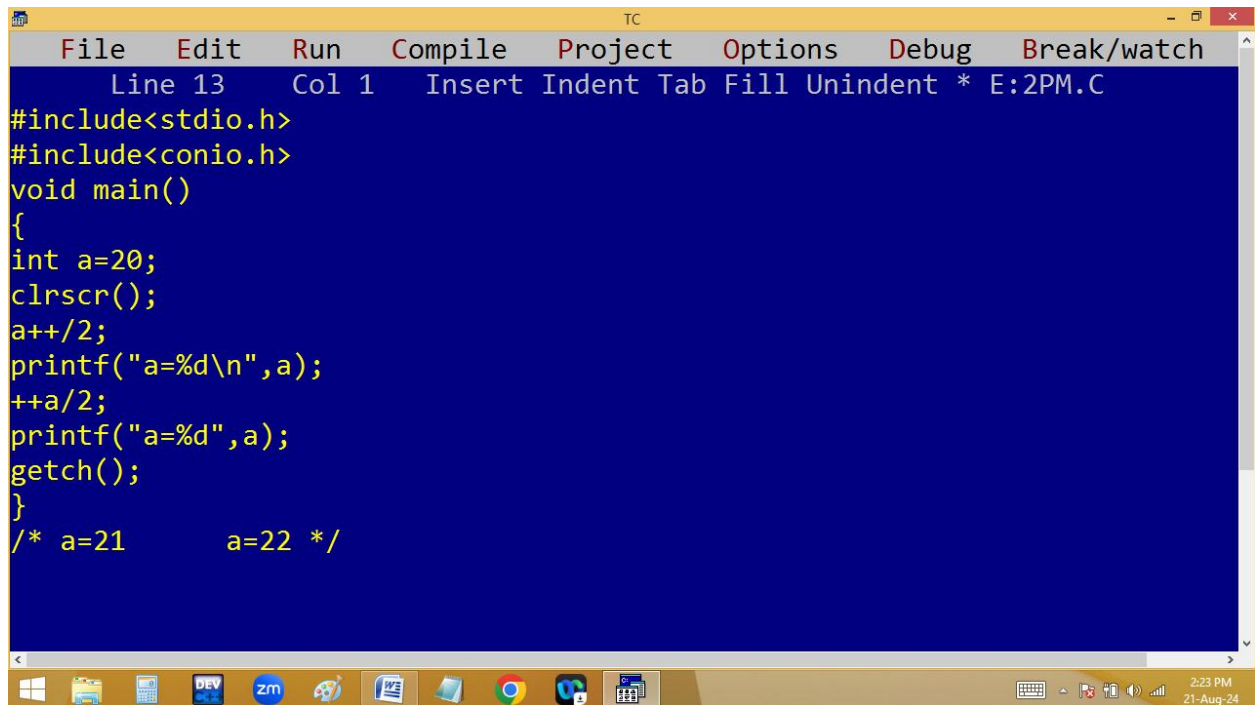




```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 13 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=20;
clrscr();
a++/2;
printf("a=%d\n",a);
++a/2;
printf("a=%d",a);
getch();
}
/* a=21 a=22 */
```

a=20

a++/2; priority: /, a++

1. a/2 ==> 20/2=10 [10 not stored in a because of = not used i.e. a=20]

2. a++ ==> a=21

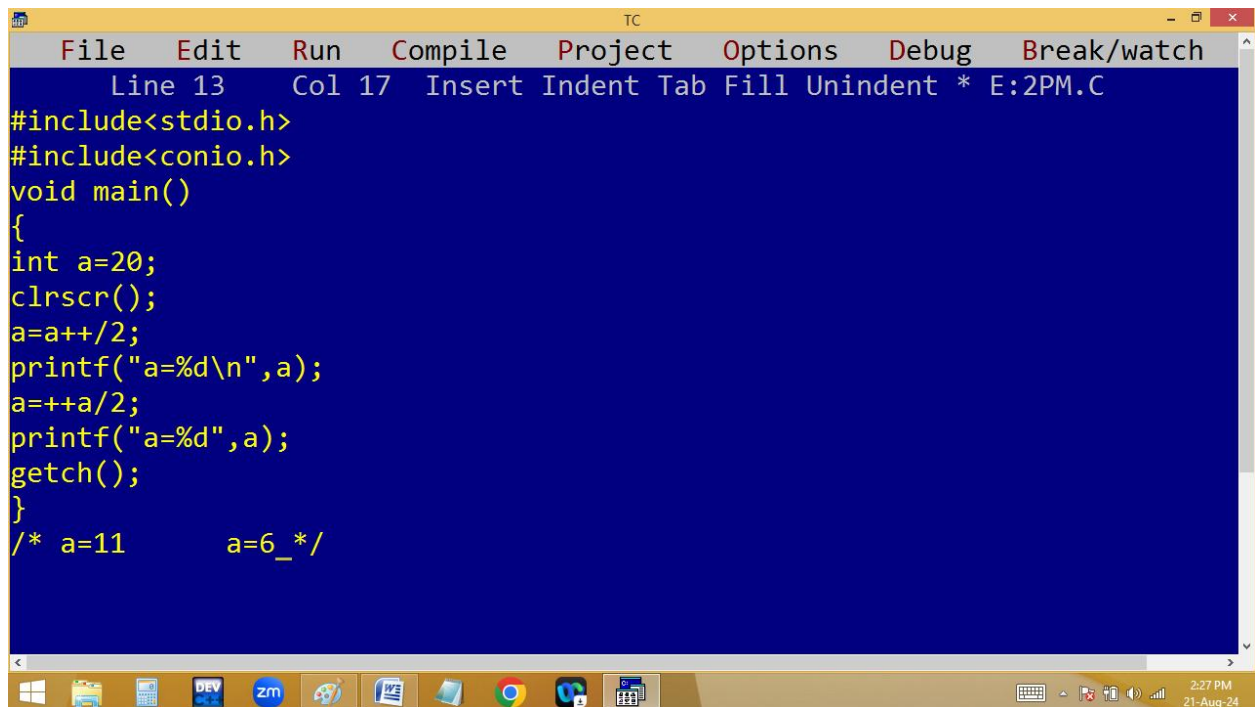
p(a) ==> 21

++a/2; priority: ++a,a/2

1. ++a ==> a=22

2. a/2 ==> 22/2 = 11 [11 not stored in a because of = not used i.e. a=22

p(a) ==> 22



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 13 Col 17 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=20;
clrscr();
a=a++/2;
printf("a=%d\n",a);
a=++a/2;
printf("a=%d",a);
getch();
}
/* a=11 a=6_*/
```

a=20

a=a++/2; priority: /, =, a++

1. a=a/2 ==> 20/2=10

2. a=10

3. a++ ==> a=11

p(a) ==> 11

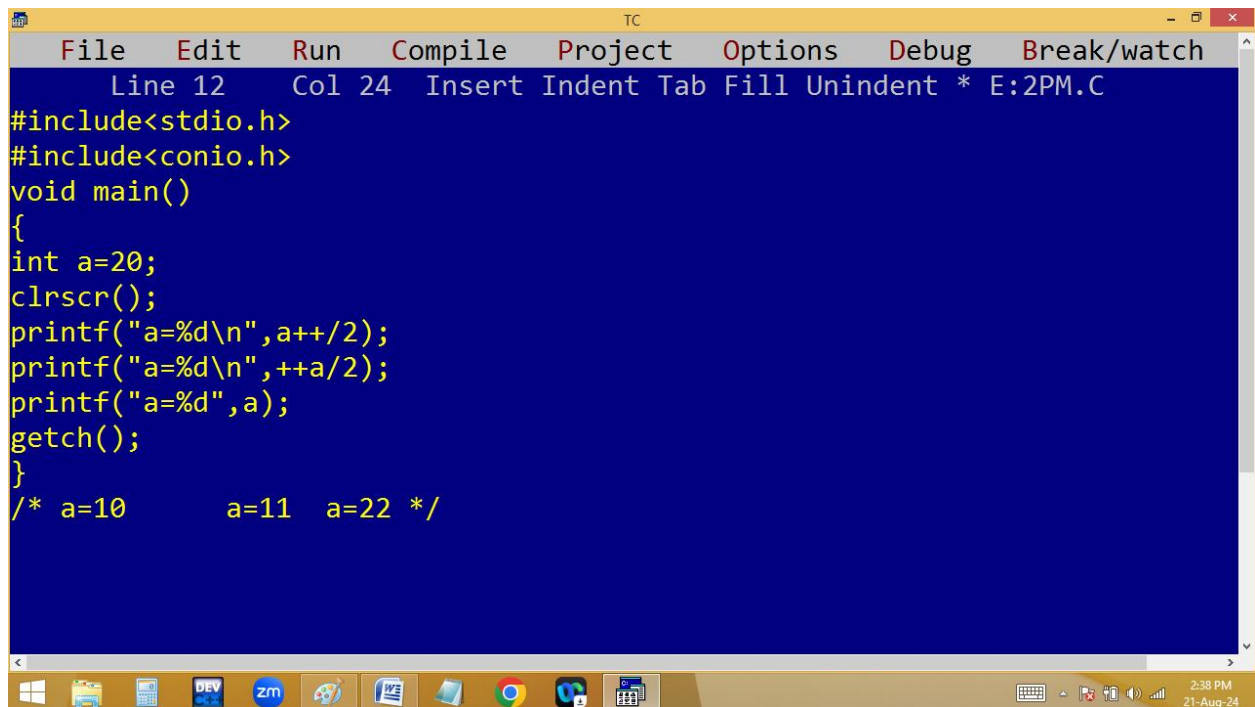
a=++a/2; priority: ++a, /, =

1. ++a ==> a=12

2. a=a/2 ==> 12/2 ==> 6

3. a=6

p(a) ==> 6



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 24 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=20;
clrscr();
printf("a=%d\\n",a++/2);
printf("a=%d\\n",++a/2);
printf("a=%d",a);
getch();
}
/* a=10      a=11  a=22 */
```

a=20

p("a=%d\\n",a++/2); priority: /, printf, a++

1. a/2==> 20/2 ==> 10

2. printf(10) 10 printed not stored i.e. **a is still 20**

3. a++ ==> a=21

p("a=%d", ++a/2); priority: ++a, /, printf

1. ++a ==> a=22

2. a/2 ==> 22/2=11

3. printf(11) ==> 11 printed not stored i.e. a=22

printf(a) ==> a=22

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 16 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=1, b=2;
clrscr();
a = a++ + ++b;
b = ++a + b++;
printf("a=%d, b=%d\n",a,b);
getch();
}
/* a=6, b=10 */
```

a=1 b=2

a = a++ + ++b; priority: ++b, +, =, a++

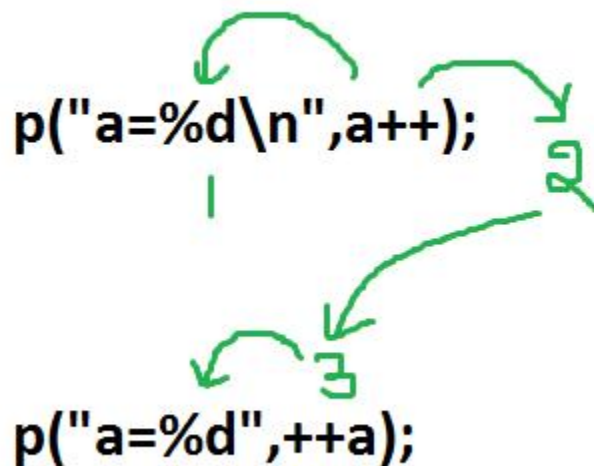
1. **++b ==> b=3**
2. **a = a + b ==> 1 + 3**
3. **a = 4**
4. **a++ ==> a=5**

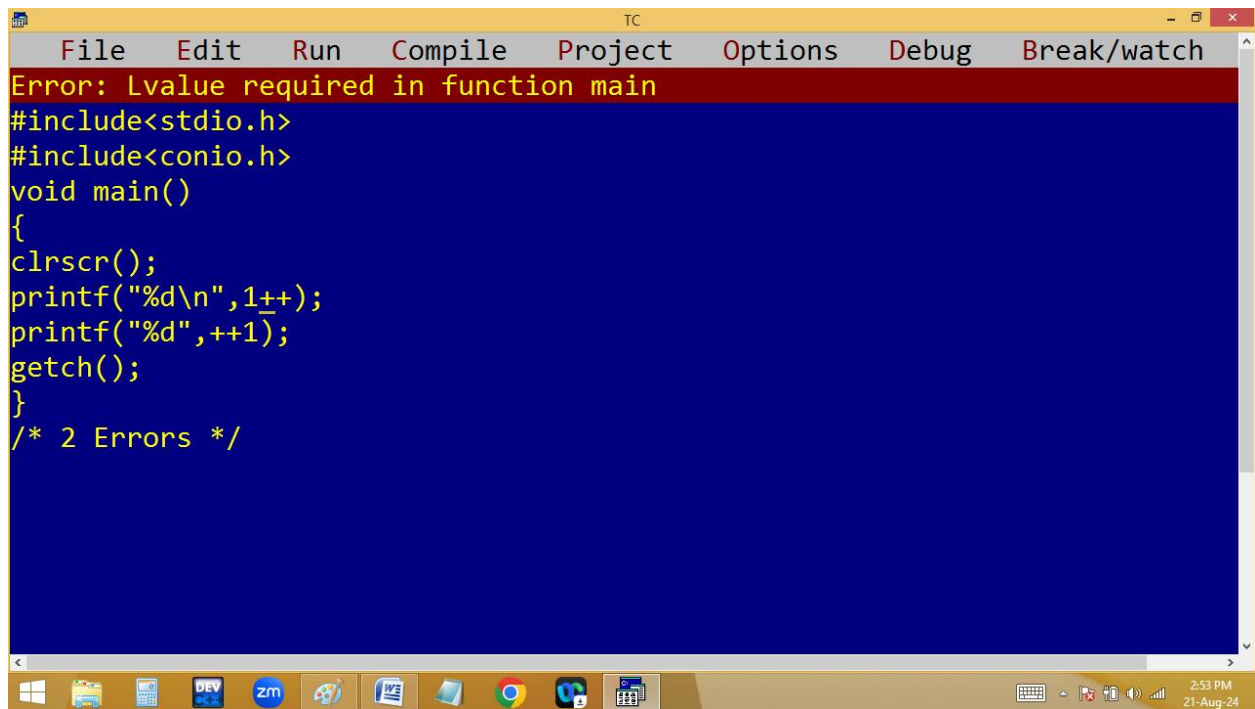
b = ++a + b++; priority: ++a, +, =, b++

1. **++a ==> a=6 ✓**
2. **b = a + b ==> 6 + 3**
3. **b = 9**
4. **b++ ==> b=10 ✓**

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 12 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=1;
clrscr();
printf("a=%d\n",a++);
printf("a=%d",++a);
getch();
}
/* a=1, a=3_*/
```

a=1





The screenshot shows the Turbo C++ (TC) IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a toolbar. The main window has a blue background with yellow text. A red error message bar at the top reads "Error: Lvalue required in function main". The code in the editor is as follows:

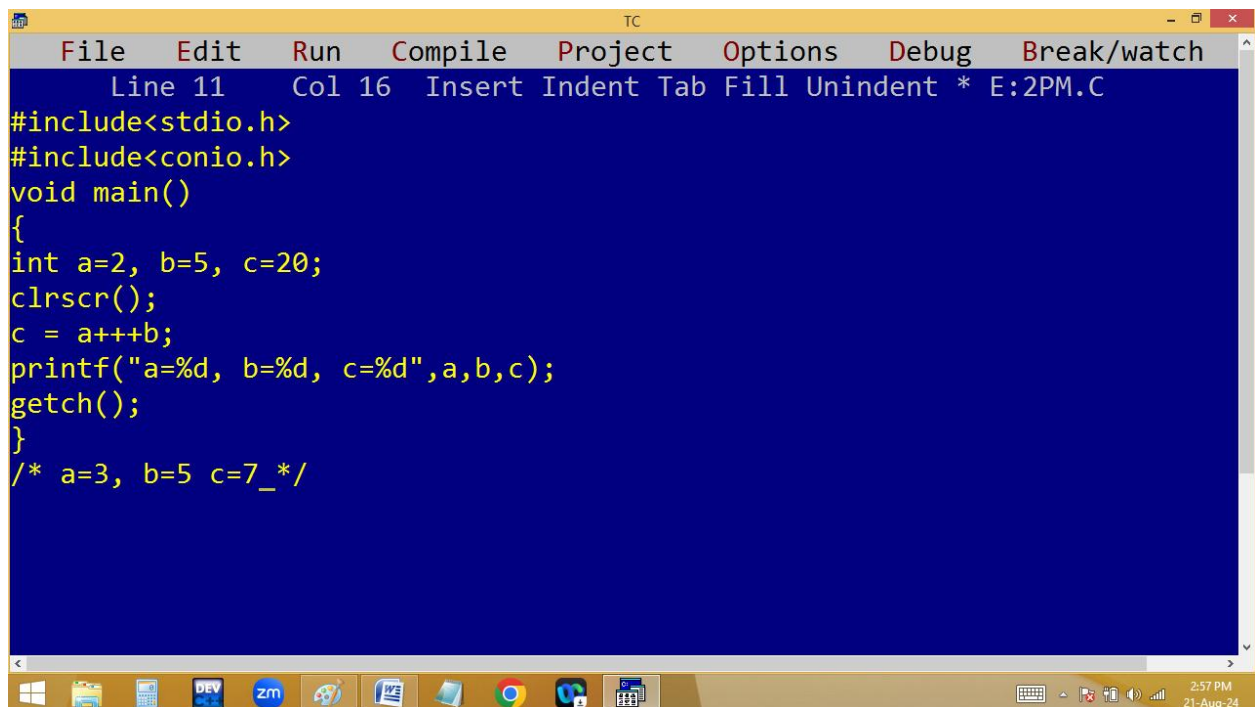
```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",1++);
printf("%d",++1);
getch();
}
/* 2 Errors */
```

The Windows taskbar at the bottom shows various icons including Windows, File Explorer, DEV, zm, and others, along with the system clock showing 2:53 PM on 21-Aug-24.

a++ means a=a+1

1++ means 1=1+1 → 1=2

Here 1 and 2 both are constants. Hence it gives errors



The screenshot shows the Turbo C++ (TC) IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a toolbar. The main window has a blue background with yellow text. A status bar at the top of the editor shows "Line 11 Col 16 Insert Indent Tab Fill Unindent * E:2PM.C". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=2, b=5, c=20;
clrscr();
c = a+++b;
printf("a=%d, b=%d, c=%d",a,b,c);
getch();
}
/* a=3, b=5 c=7_*/
```

The Windows taskbar at the bottom shows various icons including Windows, File Explorer, DEV, zm, and others, along with the system clock showing 2:57 PM on 21-Aug-24.

a=2 b=5 c=20

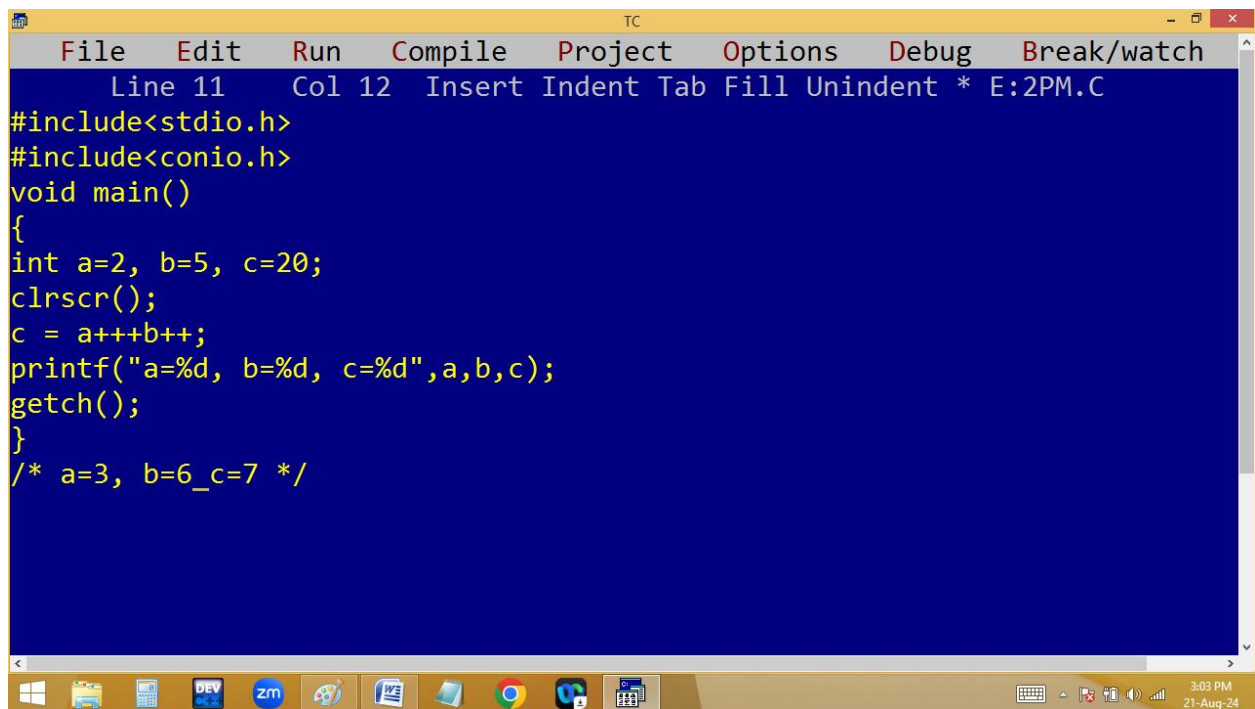
c = a++ + b;

priority: +, =, a++

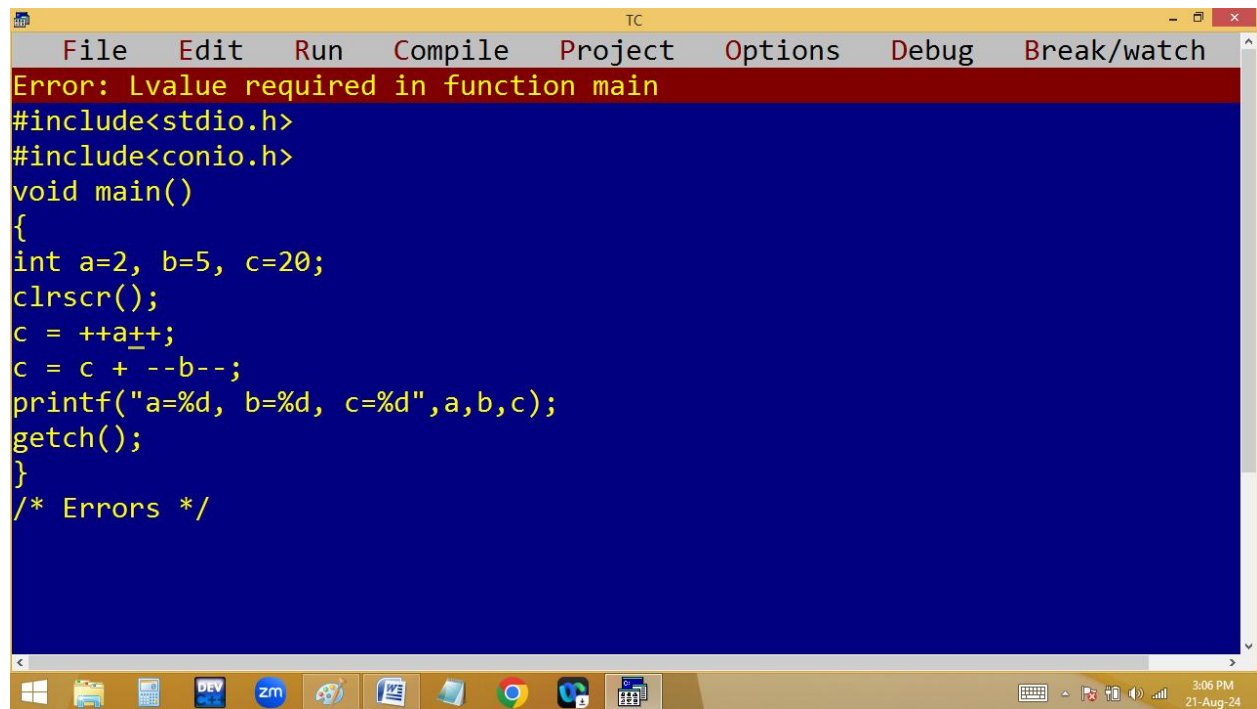
1. c = a + b ==> 2 + 5

2. c = 7

3. a++ ==> a=3



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 12 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=2, b=5, c=20;
clrscr();
c = a++ + b++;
printf("a=%d, b=%d, c=%d",a,b,c);
getch();
}
/* a=3, b=6_c=7 */
```



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", "Debug", and "Break/watch". A red error message banner at the top reads "Error: Lvalue required in function main". The code editor has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=2, b=5, c=20;
clrscr();
c = ++a++;
c = c + --b--;
printf("a=%d, b=%d, c=%d",a,b,c);
getch();
}
/* Errors */
```

The Windows taskbar is visible at the bottom, showing icons for Windows, File Explorer, Task View, DEV C++, zm, a game controller, Word, a folder, Chrome, and a terminal. The system tray on the right shows the time as 3:06 PM and the date as 21-Aug-24.

```
TC
File Edit Run Compile Project Options Debug Break/watch
Error: Lvalue required in function main
#include<stdio.h>
#include<conio.h>
void main()
{
int a=2, b=5, c=20;
clrscr();
c = (a++ + ++b)++;
printf("a=%d, b=%d, c=%d",a,b,c);
getch();
}
/* Error */
```

a=2 b=5 c=20

$c = (\overset{\uparrow 3}{a}++ + \underline{++b})++;$

$\underline{2} + 6$

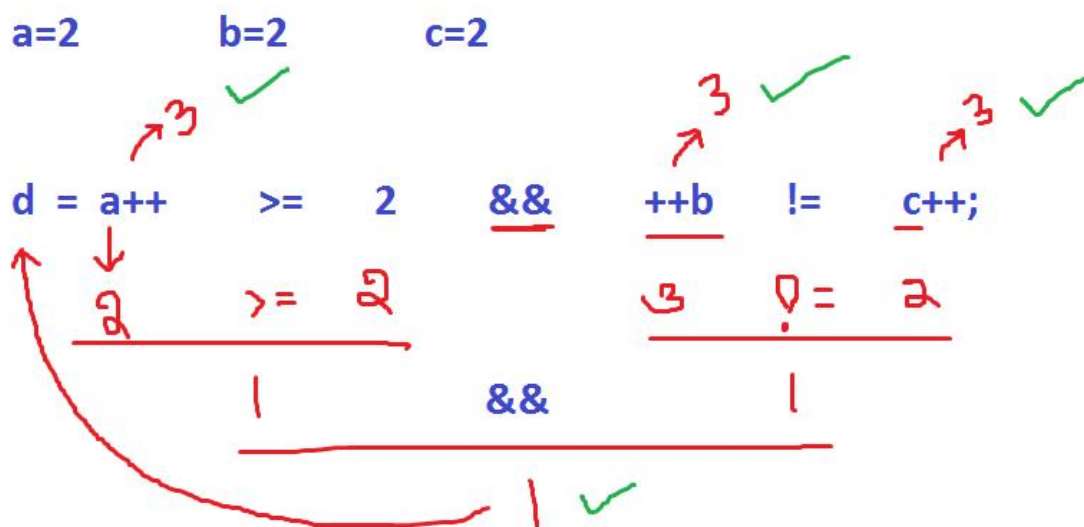
$8++$



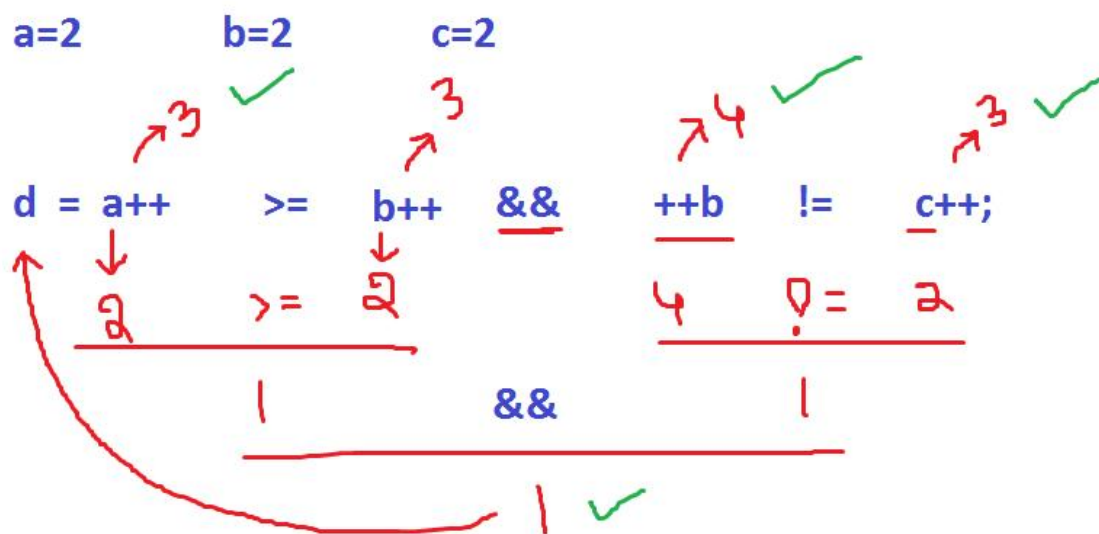
8 is a constant

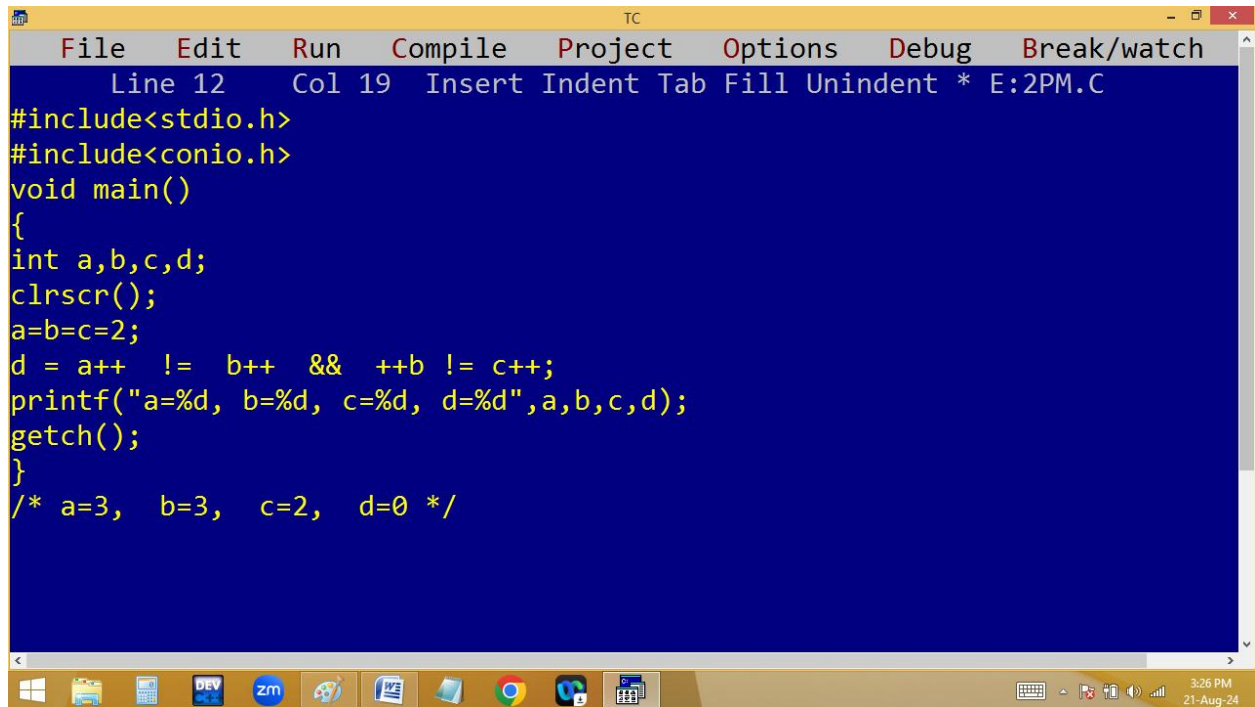
Note: We can't use ++ / -- on expressions. because of every exp final answer is constant

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 25 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b,c,d;
clrscr();
a=b=c=2;
d = a++ >= 2 && ++b != c++;
printf("a=%d, b=%d, c=%d, d=%d",a,b,c,d);
getch();
}
/* a=3, b=3, c=3, d=1 */
```

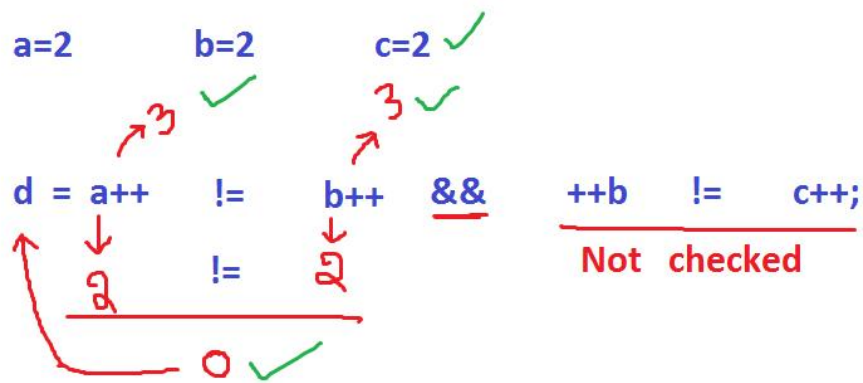


```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 13 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b,c,d;
clrscr();
a=b=c=2;
d = a++ >= b++ && ++b != c++;
printf("a=%d, b=%d, c=%d, d=%d",a,b,c,d);
getch();
}
/* a=3, b=4, c=3, d=1 */
```



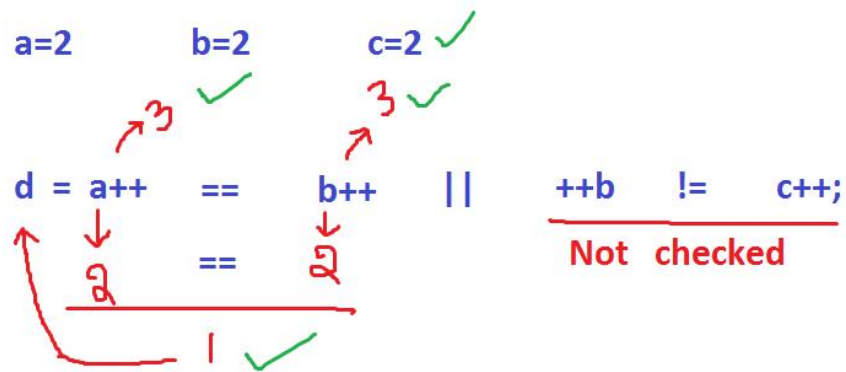


```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 19 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b,c,d;
clrscr();
a=b=c=2;
d = a++ != b++ && ++b != c++;
printf("a=%d, b=%d, c=%d, d=%d",a,b,c,d);
getch();
}
/* a=3, b=3, c=2, d=0 */
```



Note: In `&&` operation when left exp false then right exp not checked.

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 25 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b,c,d;
clrscr();
a=b=c=2;
d = a++ == b++ || ++b != c++;
printf("a=%d, b=%d, c=%d, d=%d",a,b,c,d);
getch();
}
/* a=3, b=3, c=2, d=1_*/
```



Note: In `||` operation when left exp true then right exp not checked.

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 13 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0,b=1,c=2,d;
clrscr();
d = a++ || ++b || --c;
printf("a=%d, b=%d, c=%d, d=%d",a,b,c,d);
getch();
}
/* a=1, b=2, c=2, d=1 */
```

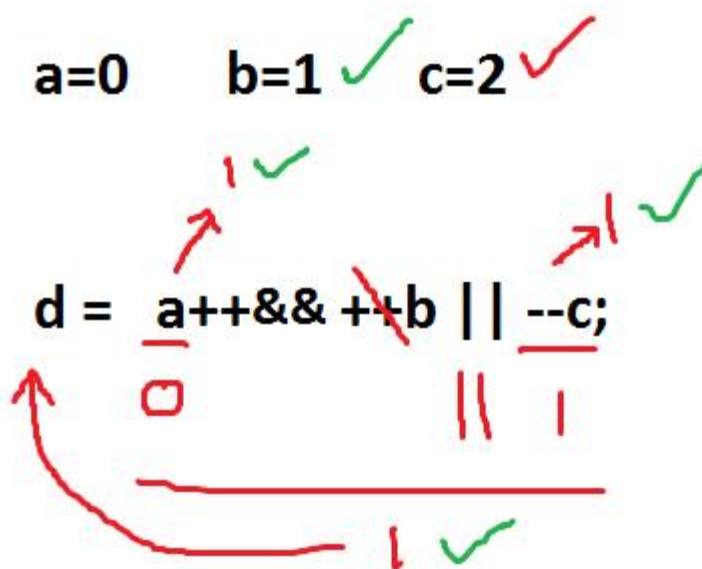
a=0 b=1 c=2 ✓

 1✓ 2✓
 ↑ ↘
d = a++ || ++b || --c;

0 || 1

1✓ || Not checked

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 21 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0,b=1,c=2,d;
clrscr();
d = a++ && ++b || --c;
printf("a=%d, b=%d, c=%d, d=%d",a,b,c,d);
getch();
}
/* a=1, b=1, c=1, d=1 */
```



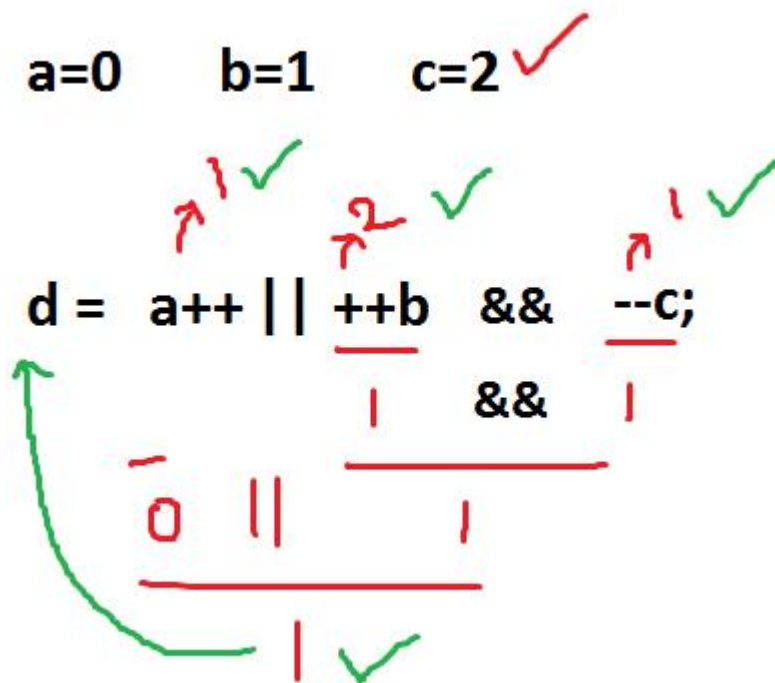
```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 25 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0,b=1,c=2,d;
clrscr();
d = a++ && ++b && --c;
printf("a=%d, b=%d, c=%d, d=%d",a,b,c,d);
getch();
}
/* a=1, b=1, c=2, d=0 */
```

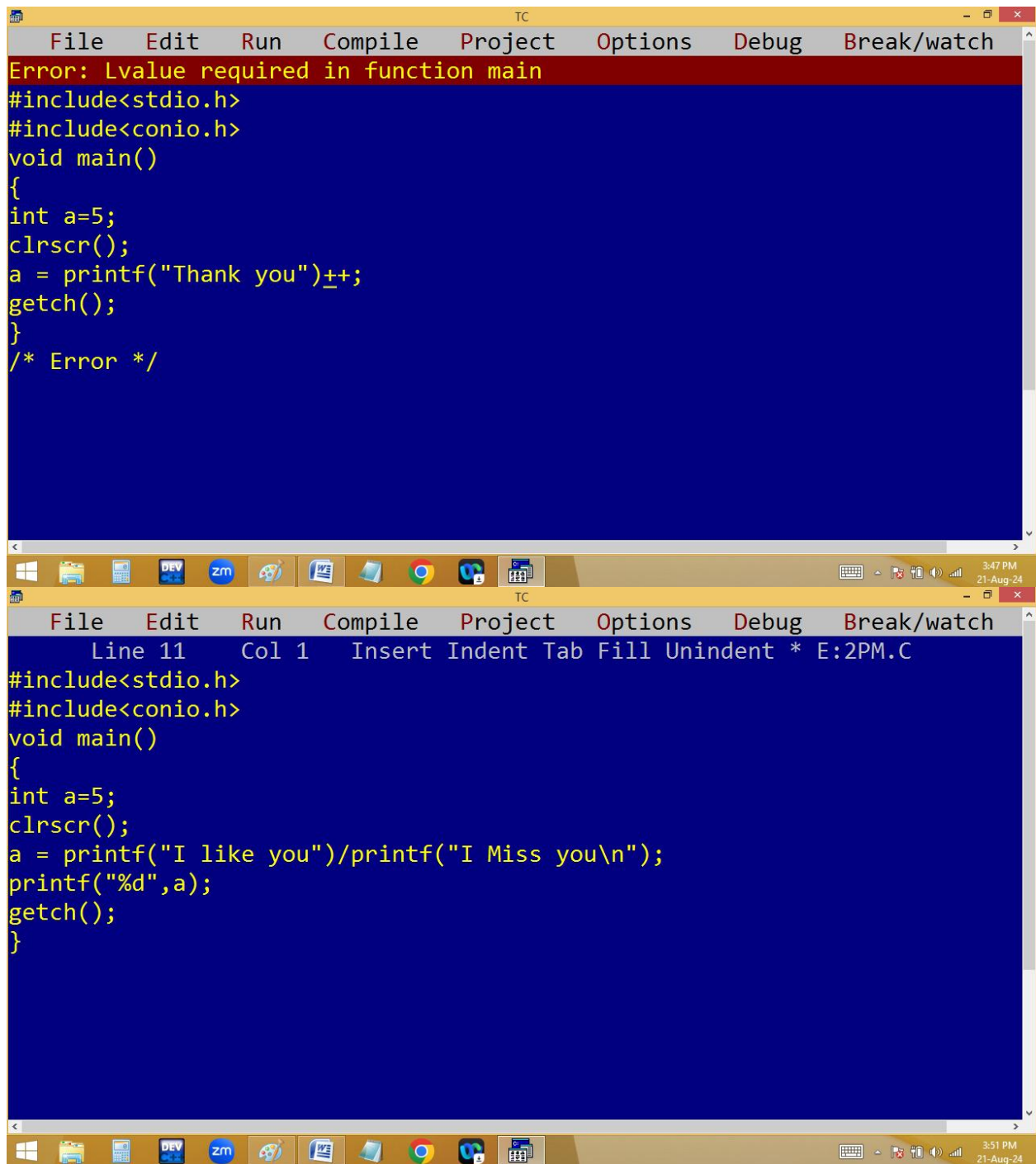
a=0 b=1 ✓ c=2 ✓

d = a++ && ~~++b~~ && ~~--c~~;

↑ ✓
↑ ✓

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 25 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0,b=1,c=2,d;
clrscr();
d = a++ || ++b && --c;
printf("a=%d, b=%d, c=%d, d=%d",a,b,c,d);
getch();
}
/* a=1, b=2, c=1, d=1_*/
```





The image shows two screenshots of the Turbo C++ (TC) IDE. The top screenshot displays a C program with a compilation error. The code is as follows:

```
File Edit Run Compile Project Options Debug Break/watch
Error: Lvalue required in function main
#include<stdio.h>
#include<conio.h>
void main()
{
int a=5;
clrscr();
a = printf("Thank you")_++;
getch();
}
/* Error */
```

The error message "Error: Lvalue required in function main" is highlighted in red. The code attempts to increment the return value of `printf` using `_++`, which is invalid. The bottom screenshot shows the same code after modification, with the error resolved. The code is:

```
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=5;
clrscr();
a = printf("I like you");printf("I Miss you\n");
printf("%d",a);
getch();
}
```

The code now prints "I like you" and "I Miss you\n" on the same line, followed by the value of `a` (5). The status bar at the bottom of the window indicates "Line 11 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C". The system tray shows the time as 3:47 PM on 21-Aug-24.

